Is space exploration a necessity or an indulgence?

Introduction

Since Neil Armstrong first stepped on the soft pale surface of the Moon, humankind has continued along the path to explore more and more about the mysterious dark space around us. Currently, humankind has managed to travel as far as 406,771 kilometers from Earth [1]; most advanced technological companies have made communication between astronauts and Earth more efficient, reduced the cost of exploration through reusable technologies, and made long-term living in space possible. Novel and unprecedented discoveries have also been frequently made by astronauts and scientists, including new organics and astonishing phenomena [2]. However, data show that NASA spent $24.8 billion on space exploration in 2025 [3], and this amount is still increasing over time. Considering the enormous cost, the progress achieved so far, and the limited direct benefits to ordinary people, a critical doubt arises: is space exploration truly necessary today, or is it pursued for other purposes, such as political prestige, commercial profit, and technological ambition?

Since current technology limits the distance and depth that humans can explore, it is hardly possible for us to find living organisms or necessary materials for human survival on other planets, indicating that establishing a new home for humankind is not an achievable short-term goal of space exploration at present. According to NASA, the primary purpose of space exploration is “(to address) fundamental questions about our place in the universe and the history of our solar system.” [3] It is admitted that space exploration does bring benefits. However, many of them are indirect and difficult to translate into immediate improvements in people's everyday lives. Nevertheless, such achievements can only truly fulfill civilizations when they are advanced enough and have already satisfied their fundamental needs [4]. “Around 733 million people faced hunger in 2023,” according to the World Health Organization (WHO) [5]. Estimates from the United Nations (UN) show that it would take approximately $93 billion per year to solve world hunger by 2030 [6]. If civilizations on Earth have not yet addressed their fundamental needs, why should they spend so much money fulfilling such an advanced need, which doesn’t even bring a clear future for human beings?

This essay argues that space exploration is not as necessary as addressing problems of hunger, disease, and poverty today. Although space exploration helps humans better understand their surroundings, facilitates the development of advanced technologies using outer-space materials, and contributes to discovering the history of planets, it would be far better to allocate the money toward addressing existing fundamental issues, considering the enormous cost of space exploration.

Comparison between two pathways

Based on established online data, it is possible to make some predictions about the future of space exploration. The total amount spent on space exploration by global governments each year is about 135 billion U.S. dollars [7], and this amount is likely to increase due to the fact that many robotic missions are gradually being transformed into human missions, indicating that the cost will continue to rise. Many achievements from space exploration remain uncertain and largely predictive in nature. Moreover, the exploration of space seems to be moving towards targets beyond revealing the history of Earth or ensuring the survival of human beings. Private enterprise in outer space appears to be an emerging and significant development for capitalism. “NewSpace does not benefit humankind as such but rather a specific set of wealthy entrepreneurs” according to Victor L. Shammas, et al [8]. This enlightening article suggests that many of the benefits of modern space exploration are becoming increasingly concentrated among governments, corporations, and wealthy individuals rather than ordinary citizens. Tourism to outer space has also emerged in recent years as space technology has developed sufficiently for human travel [9]. In the paper, the authors state that “the development of outer space has significant implications for Earth's inhabitants, yet only a small cadre of individuals, companies, and governments are involved in this process.” [9] It is apparent that the pathway of space exploration will focus on small groups of wealthy individuals. The billionaire space race is also popular today [10]. It is a competition among three famous wealthy men: Elon Musk, Jeff Bezos, and Richard Branson. They operate their companies for different purposes, but their fundamental aim is to create a new and exciting business platform for rich people around the world [10]. This race has been widely criticized because it is extremely costly and mainly serves the wealthy [10]. From the examples above, it can be argued that the future of space exploration increasingly involves business, tourism, and entertainment that mainly serve rich individuals. Enterprise and governments spend a great deal of money exploring space without average citizens being significantly involved or benefiting.

On the other hand, the dispensability of space exploration can be highlighted by comparing it with efforts to solve fundamental issues on Earth using the money originally allocated for space exploration. If humankind begins to reduce most space exploration spending and save the money, there would be an additional 135 billion dollars or more each year available for solving issues such as hunger, disease, and poverty [7, 11]. The estimated cost to end world hunger is 93 billion dollars per year [6], the economic loss caused by diseases in Africa is 2.4 trillion dollars [12], and it would take 318 billion dollars per year to end extreme global poverty [13]. The comparison of these figures shows that it is implausible to solve all these problems simply by reducing the budget for space exploration. Nonetheless, it is also obvious that if humankind starts to reduce unnecessary budget like excessive investment in space exploration, millions of people suffering from fundamental issues could be saved or their living conditions could be slightly improved.

This comparison does not mean that scientific research is unimportant. However, it raises a question about what societies should focus on first. If resources are limited, is it more reasonable to prioritize the development of cutting-edge technologies with uncertain and uneven returns, or to use those resources to address fundamental issues such as hunger, disease, and poverty? Given the challenges that still exist today, this essay supports the latter view.

History of space exploration

By briefly describing the previous steps of space exploration, patterns are revealed. The Soviet Union launched the first artificial Earth satellite in 1957. The first applications satellite was launched by the United States in 1960. The Soviet Union sent the first human into orbit around the Earth in 1961. The United States obtained data from another planet for the first time in 1962. The Soviet Union completed the first spacewalk in 1965. The United States and the Soviet Union cooperated to construct the first international docking in space in 1975. The first spacecraft to make a close flyby of a comet nucleus was launched by the European Space Agency in 1986. China achieved the first landing on the far side of the Moon in 2019. Many vital achievements are not listed here.

Most of these achievements took place during the Cold War between the United States and the Soviet Union. It suggests that the major purpose of developing space technology and exploring space went beyond revealing mysteries and understanding the history of the universe. The primary aim was to compete against another powerful country and demonstrate national strength. In pursuing this goal, the Soviet government devoted substantial resources to advanced technology and heavy industry while neglecting the development of the fundamental needs of the country. Generally, there were more disadvantages than benefits in such a situation.

Unnecessary alternative purpose of space exploration

Analyzing the situation today, it is obvious that humanity is still stepping on this path that overlooks the fundamental issues of civilizations. Governments and technology companies like SpaceX keep on investing tons of money into space exploration and advancing their tools, while issues such as hunger, disease, and poverty remain unresolved.

Public opinion data further supports this concern. A large survey conducted in 2022 regarding the attitude of American citizens toward space exploration reveals supporting data across various time periods. During the period from 1961 to 1970, the proportion of American citizens opposed to space exploration increased from below 20% to over 60% [14]. When respondents were asked whether $4 billion per year should be

spent on the space program or redirected toward areas such as education, desalination of water, or reducing the national debt, between 55% and 60% regarded these alternatives as more important than space exploration [14]. Furthermore, when asked what the United States should prioritize over the following two years, fewer than 5% selected space exploration [14]. Although 72 percent of American respondents in 2018 believe it was “essential” for America to continue being a world leader in space exploration, public concern regarding the scale of spending has remained consistently high [14]. From 1975 to 2021, the number of people who believed the budget spent on space exploration was too high consistently remained greater than the proportion who thought the spending was appropriate or too low, regardless of political affiliation [14]. The conclusion of the survey stated that “they (advocates for space exploration) will continue to have their work cut out for them for the foreseeable future.”

This distinction holds significant importance at the philosophical level. Many citizens do not oppose scientific exploration; they merely question whether such exploration should take precedence over those urgent social and economic issues that directly affect daily life and require immediate resolution. In the current global situation, if millions of people still cannot obtain basic necessities of life, then the massive investment in projects mainly related to prestige, competition, and the interests of the elite seems hardly justifiable.

Conclusion

The reason for the massive achievements of the Industrial Revolutions was the existence of sufficient fundamental support and resources. It is true that space exploration brings humanity surprising discoveries comparable to those produced during each Industrial Revolution. However, the purpose of space exploration is not necessarily to achieve higher efficiency, higher average income, or better living quality. Space exploration is increasingly influenced by considerations of profit, fulfilling the rich, and national prestige. With so many serious fundamental issues related to human survival continuing to worsen over time, it is difficult to regard space exploration as a necessity. Instead, it appears closer to an indulgence, and it is more important to direct resources toward addressing these fundamental challenges for the healthy development of human societies. In this sense, the debate over space exploration is ultimately a debate about priorities.

Furthermore, other areas of government expenditure, particularly military spending, could contribute more directly to addressing problems such as hunger, considering that ending world hunger costs less than 1% of global military spending [6]. However, the prior condition for this is the existence of world peace and freedom. In such a situation, excessive spending on technological competition and national defense would no longer be necessary, indicating that the fundamental issues mentioned above could be alleviated or even

solved.

References

[1]. Iain Todd, They've done it. The Artemis II crew have travelled further into space than any human being before them, Astronomy News, Published: April 7, 2026. https://www.skyatnightmagazine.com/news/artemis-ii-crew-record-breaking-distance-from-earth

[2]. Amy J. Williams, Jennifer L. Eigenbrode, Maëva Millan, et al. Diverse organic molecules on Mars revealed by the first SAM TMAH experiment. Nat Communications 17, 2748 (2026). https://www.nature.com/articles/s41467-026-70656-0

[3]. https://www.nasa.gov/

[4]. Key components of civilization, National Geographic, Education. https://education.nationalgeographic.org/resource/key-components-civilization/

[5]. Saving lives changing lives, World Food Programme. https://www.wfp.org/support-us

[6]. Ending world hunger costs less than 1% of military spending, UN News. https://news.un.org/en/story/2025/11/1166397

[7]. Government expenditure on space programs in 2022 and 2024, by major country (in billion U.S. dollars). https://www.statista.com/statistics/745717/global-governmental-spending-on-space-programs-leading-countries/

[8]. Victor L. Shammas, Tomas B. Holen, One giant leap for capitalistkind: private enterprise in outer space. Palgrave Communications 5, 10 (2019). https://www.nature.com/articles/s41599-019-0218-9

[9]. Sam Spector and James E.S. Higham, Space tourism in the Anthropocene, Annals of Tourism Research, 79, 102772 (2019). https://www.sciencedirect.com/science/article/abs/pii/S016073831930129X

[10]. Billionaire space race. https://en.wikipedia.org/wiki/Billionaire_space_race

[11]. A heavy burden: the productivity cost of illness in Africa. https://www.afro.who.int/publications/heavy-burden-productivity-cost-illness-africa

[12]. Diseases cost Africa $2.4 trillion per year, WHO reports, Global Health Matters May/Jun 2019. https://www.fic.nih.gov/News/GlobalHealthMatters/may-june-2019/Pages/who-report-productivity-cost-illness-in-africa.aspx

[13]. Diccon Hyatt, How much would it cost to end extreme poverty worldwide? April 16, 2026, https://www.investopedia.com/how-much-would-it-cost-to-end-extreme-poverty-worldwide-11946764

[14]. John Konicki and James Pethokoukis, Do Americans care about space? The New Atlantis 68, 90-108 (2022). https://www.thenewatlantis.com/publications/do-americans-care-about-space